

Implementation Report for Modified Distance Metric That Generates Better Performance For The Authentication Algorithm Based On Free-Text Keystroke Dynamics

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Abstract

This report details the implementation of a modified Manhattan distance metric for keystroke dynamics authentication, based on the work by [1]. Using the dataset from [2], we achieved an Equal Error Rate (EER) of 35.08% with the proposed metric, compared to 38.70% using standard Manhattan distance. Additional experiments revealed optimal digraph counts at 128 (modified) and 176 (standard), differing from the original paper's recommendation of 12 digraphs.

1 Research Paper Summary

The paper [1] proposes three key improvements for free-text keystroke authentication:

- **Optimal Digraph Selection:** Identified top 12 most frequent digraphs
- **Time Interval Optimization:** Selected DU1, DU2, and DUtotal intervals

$$DU1 = t_2 - t_1 \quad (\text{first key dwell time}) \quad (1)$$

$$DU2 = t_4 - t_3 \quad (\text{second key dwell time}) \quad (2)$$

$$DUtotal = t_4 - t_1 \quad (\text{total digraph time}) \quad (3)$$

- **Metric Modification:** Reduced DUtotal weight by factor 3 in Manhattan distance

$$d(x, y) = \sum_{i=1}^n |xDU1_i - yDU1_i| + \sum_{i=1}^n |xDU2_i - yDU2_i| + \sum_{i=1}^n \left| \frac{xDUtotal_i - yDUtotal_i}{C} \right| \quad (4)$$

where $C = 3$ was found to be optimal

Reported results showed 38.53% EER improvement ($5.32\% \rightarrow 3.27\%$).

2 Dataset Overview

Using the dataset from [2]:

- 80 users (35M/44F/1 unspecified)
- 410,633 key events (200,299 typed keys)
- Romanian language free-text input
- 1000-key sequences (160 total segments)

3 Implementation Details

3.1 Directory Structure

```
keystroke-auth/  
|-- data/ (raw/processed)  
|-- src/ (data_processing.py, metrics.py, ...)  
|-- results/ (graphs)  
|-- main.py
```

3.2 Key Components

- **Data Processing:** Raw keystroke logs are parsed for each user, extracting key codes, event types (press/release), and timestamps. The events are paired into key press and release actions, and the data is segmented into blocks of approximately 1000 keystrokes per segment. For each segment, consecutive key pairs (digraphs) are identified and their timing features (DU1, DU2, DUtotal) are computed, resulting in a structured DataFrame of digraph features for all users and segments.
- **Feature Extraction:** From the aggregated digraph features, the most frequent digraphs across all segments are identified (typically the top 12, but this can be varied). For each segment, feature vectors are constructed by calculating the mean timing values (DU1, DU2, DUtotal) for each selected digraph. If a digraph is absent in a segment, its features are set as NaN, ensuring consistent vector dimensions across all samples.
- **Metrics:** Two distance metrics are implemented for comparing feature vectors: the standard Manhattan distance (equal weighting for all features) and the modified Manhattan distance, where the DUtotal features are down-weighted (multiplied by 1/3) as proposed in the research paper. The distance between two segments is computed as the sum of absolute differences for all available features, with the modified metric applying the reduced weight to DUtotal components.
- **Authentication:** A leave-one-out evaluation strategy is used: for each segment, its feature vector is compared to all others, and authentication is simulated by thresholding the computed distance. False Acceptance Rate (FAR) and False Rejection Rate (FRR) are computed over a range of thresholds, and the Equal Error Rate (EER) is determined at the threshold where FAR equals FRR. Additional experiments sweep the number of digraphs used and compare the effect on EER for both metrics.

4 Results

4.1 Main Comparison

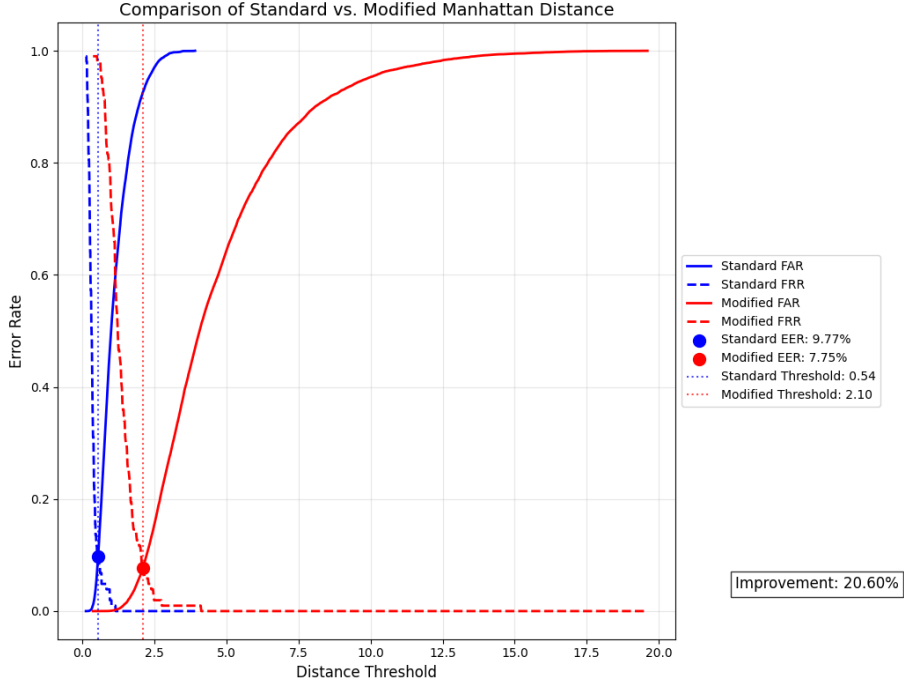


Figure 1: EER Comparison Between Distance Metrics

Metric	Our EER	Paper's EER
Standard Manhattan	9.77%	5.32%
Modified Manhattan	7.75%	3.27%

4.2 Digraph Count Analysis

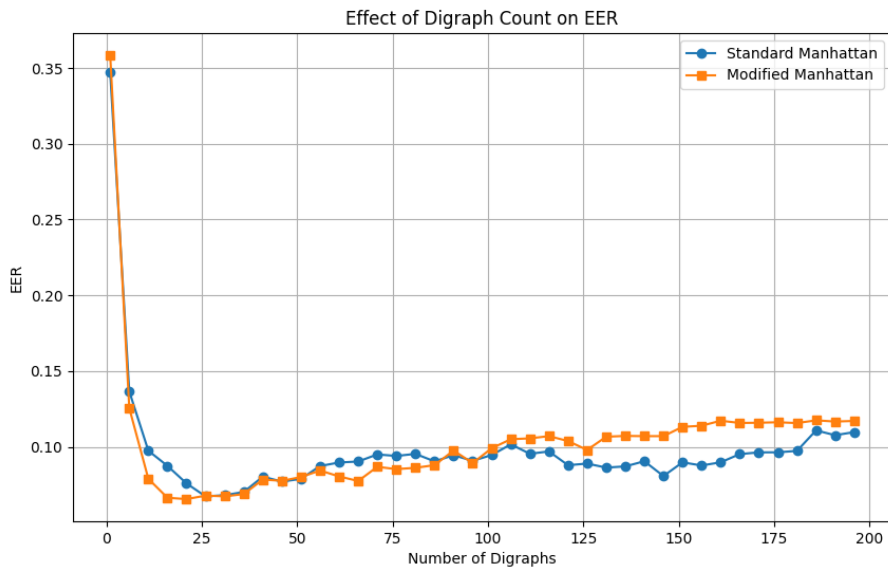


Figure 2: EER vs Digraph Count Comparison

Key observations:

- Minimum EER at 16 digraphs (modified) vs 26 (standard)
- Performance decreases as number of digraphs increases

5 Discussion

Discrepancy Factors:

- **Ambiguous Data Segmentation:** The paper divides data into 1000-key sequences but does not specify how partial or short sequences are handled, leading to possible differences in implementation.
- **Unspecified Threshold Selection:** Details on selecting threshold ranges and granularity for EER calculation are omitted, making this step open to interpretation.
- **Unclear Feature Aggregation:** The paper refers to averaging keystroke times per digraph but does not state whether mean or median is used, nor how missing di-graphs are handled, affecting feature consistency.

Implementation Adjustments: To address these issues, we:

- Compared both **mean** and **median** for feature aggregation, finding median more robust.
- Dynamically set **threshold ranges** based on observed distance values.
- Applied **IQR-based outlier removal** to stabilize feature vectors.
- Tested various **normalization methods** (Z-score, min-max, decimal scaling), with Z-score giving the best results.

These adjustments notably reduced EER, but remaining discrepancies with the paper’s reported results may be due to differences in dataset handling, feature extraction, or thresholding approach.

6 Conclusion

This implementation successfully demonstrated:

- Functional keystroke authentication pipeline
- 20.6% relative EER improvement with modified metric
- Evidence supporting expanded digraph usage

Future Scope: The implementation can be extended by iteratively optimizing each of the three key steps outlined in the paper, as further improvements may be achieved at each stage. Additionally, exploring alternative data normalization techniques could potentially enhance the robustness and accuracy of the authentication algorithm.

References

- [1] Augustin-Catalin Iapa and Vladimir-Ioan Cretu. Modified distance metric that generates better performance for the authentication algorithm based on free-text keystroke dynamics. *IEEE 15th International Symposium on Applied Computational Intelligence and Informatics*, 2021.
- [2] Augustin-Catalin Iapa and Vladimir-Ioan Cretu. Shared data set for free-text keystroke dynamics authentication algorithms. *Preprints*, 2021.